

Solano County Health & Social Services Department



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POLICY MEMORANDUM 5900

Implementation Date: April 7, 2011

Revised Date: May 1, 2017

Review Date: May 1, 2019

REVIEWED/APPROVED BY:

A handwritten signature in blue ink, appearing to read "Bryn".

BRYN E. MUMMA, MD, MAS, EMS AGENCY MEDICAL DIRECTOR

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TED SELBY, EMS AGENCY ADMINISTRATOR

SUBJECT: SOLANO COUNTY TRAUMA SYSTEM

AUTHORITY: California Health & Safety Code, Division 2.5, Section 1797.220 and 1798.163.

PURPOSE/POLICY:

This policy shall provide the basic outline of the Trauma System in Solano County including but not limited to, trauma system design, participants, Trauma Center geographic service areas, trauma data collection, analysis and management, coordination of trauma care with neighboring jurisdictions, and quality improvement and evaluation.

I. DEFINITIONS:

- A. Trauma Center:** Designation received by a hospital after undergoing extensive review from the Local EMS Agency and complying with the appropriate State of California regulations. These hospitals provide trauma data to Solano County EMS Agency.
- B. Pediatric Trauma Center:** Designation received by a hospital after undergoing extensive review from the Local EMS Agency and complying with the appropriate State of California regulations.
- C. Receiving Hospital:** A system hospital with a Basic Emergency Department as defined in the appropriate State of California regulations.

II. OVERVIEW OF SOLANO COUNTY TRAUMA SYSTEM:

Solano County is a medium sized county with a population of approximately 420,000 located in the Bay Area region of California. The county has five community hospitals and one military hospital providing care to the residents. The Solano County EMS in-county Designated Level II Trauma Center is Kaiser Foundation Hospital, Vacaville (KVV); the in-county Designated Level III Trauma Center is NorthBay Medical Center (NBMC). Solano County EMS has also designated John Muir Medical Center (JMMC), Walnut Creek and UC Davis Medical Center (UCDMC), Sacramento as out-of-county Level II and I Trauma Centers respectively. The closest pediatric trauma centers are located in Sacramento, UCDMC and Alameda Counties, Children's Hospital Oakland (CHO). Based on the Solano County Trauma Triage criteria the most severe pediatric trauma cases are transported via ground or air ambulance to the out-of-county pediatric trauma centers. Those patients who do not meet the Solano County Trauma Triage criteria are transported to a local hospital for assessment and treatment. Solano Trauma Triage criteria was developed with input from all of the constituents of our EMS system. This document is reviewed periodically by the EMS Agency Medical Director and other groups. The Physician's Advisory Forum (advisory group to the Medical Director) and Trauma Advisory Committee (TAC) provide information on quality improvement, policies, and procedures to the Solano County EMS Agency Medical Director.

A. TRAUMA CENTER CATCHMENT BY SOLANO COUNTY CITIES:

Generally trauma patients living in the following cities have this geographic distribution:

CITY	ADULTS	PEDIATRIC
Vallejo	Victims of Level I and II trauma go to JMMC	Victims of Level I and II trauma go to CHO
Benicia	Victims of Level I and II trauma go to JMMC	Victims of Level I and II trauma go to CHO
Fairfield	Victims of Level I and II trauma go to KVV or JMMC	Victims of Level I and II trauma go to CHO
Suisun City	Victims of Level I and II trauma go to KVV or JMMC	Victims of Level I and II trauma go to CHO
Rio Vista	Victims of Level I and II trauma go to KVV or UCDMC	Victims of Level I and II trauma go to UCDMC
Vacaville	Victims of Level I and II trauma go to KVV or UCDMC	Victims of Level I and II trauma go to UCDMC
Dixon	Victims of Level I and II trauma go to KVV or UCDMC	Victims of Level I and II trauma go to UCDMC

B. PHYSICIANS' ADVISORY FORUM

This is an advisory group to the EMS Agency Medical Director comprised of select individuals who provide feedback on trauma care, policies, and protocols. See Policy 1790.

C. TRAUMA ADVISORY COMMITTEE (TAC)

This committee was formerly titled Pre-hospital Trauma Advisory Committee; the group was asked to change their focus in 2013 and thus was renamed to better reflect their new role. More detailed information is provided in Section V(A)(1).

D. TRAUMA DATA COLLECTION

The Solano County EMS Agency works with our trauma center partners to collect trauma data elements pursuant to California Health & Safety Code, Sections 100257 and 100176.

III. TRAUMA CARE SYSTEMS FOR QUALITY IMPROVEMENT:

- A. Solano County Designated Trauma Centers are responsible for conducting quality improvement activities in accordance with the requirements of the various guiding documents, such as but not limited to, designation contracts and American College of Surgeons' (ACS) guidelines.
- B. Representatives from trauma centers with designations from Solano County are required to:
 - 1. Participate in the Solano County EMS Agency TAC;
 - 2. Submit trauma reports and analysis regarding patients received from Solano County to the Solano County EMS Agency as appropriate; and
 - 3. Notify the Solano County EMS Agency of unusual occurrences or other significant matters
- C. Trauma centers located in neighboring jurisdictions which are not designated by Solano County are responsible for conducting quality improvement activities in accordance with the requirements of their designation contracts.
- D. Representatives from trauma centers located in neighboring jurisdictions, which are not designated by Solano County, are invited to:
 - 1. Participate in the Solano County EMS Agency TAC;
 - 2. Submit trauma reports and analysis regarding patients received from Solano County to the Solano County EMS Agency as appropriate; and
 - 3. Notify the Solano County EMS Agency of unusual occurrences or other significant matters.

IV. GENERAL TRAUMA FIELD OPERATIONAL CONCEPTS:

- A.** Solano County Paramedics will follow the Solano County Trauma Triage Policy and Algorithm when determining a trauma patient's destination. Mode of transportation will be based on the following factors (this is not an all-inclusive list of considerations): time of day, day of week, traffic, scene location, distance to trauma center, and resource availability.
- B.** After the destination and mode of transportation decisions have been made, transport will be to the closest appropriate facility.

V. TRAUMA SYSTEM QUALITY IMPROVEMENT AND EVALUATION:

QI and evaluation of the Trauma Plan must be focused on two primary objectives: providing optimal care for trauma patients and reducing injuries through education and prevention. The trauma system quality improvement and evaluation will be done using a multi-disciplinary approach involving an improvement team comprised of individuals from local BLS and ALS providers, local receiving hospitals, and receiving trauma centers. Feedback will be directed to the appropriate individual, agency, or committee.

Evaluation parameters will include, at a minimum, measurements of trauma on-scene time and transport times, evaluation of helicopter transports, determination of over triage and under triage rates, common mechanism of injury, determination of preventable deaths, complications, average patient ages, lengths of stay, Intensive Care Unit days, and discharge status.

The Trauma Improvement Team provides quality improvement and oversight to various system agencies as data is evaluated. It will also produce generic educational materials to enhance trauma care. Because of the confidential nature of this Team, confidentiality statements are required of all participants.

A. TRAUMA ADVISORY COMMITTEE (TAC)

TAC evaluates the trauma system in Solano County. The committee evaluates the cumulative trauma data the EMS Agency collects and specific cases which have some benefit to the various providers in our system.

1. TAC COMPOSITION

The TAC shall be chaired by the Solano County EMS Agency Medical Director. The TAC membership shall include, but is not limited to:

- Solano County EMS Agency Staff
 - EMS Medical Director
 - EMS Administrator
 - EMS Associate Administrator
 - EMS Operations Manager
 - EMS Coordinator(s)
 - Other EMS Staff as directed

- Trauma Centers Designated by Solano County
 - Trauma Program Medical Director(s)
 - Trauma Program Nurse Coordinator(s)
- Solano County Designated Trauma Centers Located outside Solano County
 - Trauma Program Medical Director(s)
 - Trauma Program Nurse Coordinator(s)
- Solano County ALS Providers and Air Ambulance Providers
 - Operations Manager
 - CQI Coordinators
- Other Invited Guests as approved by the Committee chairperson

2. TAC CASE SELECTION:

Cases presented at the TAC meeting will be selected by the EMS Operations Manager and/or EMS Coordinator in consultation with the EMS Medical Director. In general, and based on the allotted time for the TAC meeting, the following types of cases will be selected: trauma death cases, patients transported to local hospital and subsequently transferred to a trauma center, patients with ISS scores of ≤ 10 and LOS of ≤ 3 , patients discharged from trauma centers in less than 24 hours.

3. TAC CONFIDENTIALITY:

- a. All proceedings, documents, and discussions of the TAC, and its subcommittees are confidential and protected under Sections 1040 and 1157.7 of the Evidence Code of the State of California. The prohibition relating to discovery or testimony provided in Section 1157 shall be applicable to proceedings and records of any committee established by a local governmental agency to monitor, evaluate, and report on the necessity, quality, and level of specialty healthcare services, including but not limited to, trauma care service provided by a general acute care hospital which has been designated or recognized by that governmental agency as qualified to render specialty healthcare services.
- b. All members and guests sign a confidentiality agreement memorializing that they will not divulge or discuss publicly information obtained through Committee membership. Prior to a guest participating in the meeting, the Committee Chair is responsible for explaining and obtaining a signed confidentiality agreement from the guest.

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Solano County Trauma Triage

Mental Status and Vital Signs	Minimum Blood Pressure by Age					
- Motor GCS < 6 (Unable to follow commands) - RR < 10 or > 29 - SpO2 < 90% on room air - Severe respiratory distress or need for support - HR above SBP consistently for patients ≥ 10 yo - SBP below minimum for age	Age	SBP	Age	SBP	Age	SBP
	0	70	4	78	8	86
	1	72	5	80	9	88
	2	74	6	82	10 - 64	90
	3	76	7	84	≥ 65	110
Injury Pattern	Mechanism of Injury					
- Penetrating injury to the head, neck, torso, or extremity at, or proximal to, the elbow or knee - Skull fracture or suspected skull fracture - Spinal injury, or suspected spinal injury, with new motor or sensory loss - Chest wall instability, deformity, or flail chest - Pelvic fracture or suspected pelvic fracture - Long bone fracture (femur or humerus), 1 for Pediatric or 2 for Adult - Extremity crushed, degloved, mangled, or pulseless - Extremity amputated at, or proximal to, the wrist or ankle - Hemorrhage requiring tourniquet application or wound packing with continuous direct pressure	- Fall > 10 ft for any age - Rider separated from transport vehicle or significant impact (motorcycle, bike, horse, etc.) - Pedestrian thrown, run over, or significant impact (i.e. struck by transport vehicle or auto) - Auto crash with high risk: - Death in same vehicle - Ejection, partial or complete - Entrapped patient with need for extrication - Intrusion of > 12 in where patient is seated - Intrusion of > 18 in into occupant compartment - Vehicle telemetry data consistent with potential for severe injury - Child < 10 yo unrestrained, or in unsecured safety seat					

Trauma Activate
Transport to closest Trauma Center, make **Trauma Base Contact** for any alternate Trauma Center transport

